

Forward passes by hand

Project: Image-pattern Network Investigation

1. Your mission

Your learning target

I can record scores and activations in dependency order.

I can locate the first incorrect value in a forward trace.

I can separate a final output from evidence of model accuracy.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Record scores and activations in dependency order

Locate the first incorrect value in a forward trace

Separate a final output from evidence of model accuracy

Retrieve what you already know

In Week 5, what does the output layer receive?

My starting explanation

This week's end goal

By the end, I can explain forward passes by hand using a traced example and a stated limitation.

2. Notice the evidence

Use a new one-input network with two hidden units. Its scores are $z_1 = 2x - 1$ and $z_2 = -x + 2$. Apply ReLU to obtain h_1 and h_2 . The output is $y = h_1 - 2h_2 + 1$. Parameters stay fixed while the input changes. The input x is an abstract number; these are arithmetic examples, not measurements of children.

A forward pass evaluates this network in dependency order. Compute both scores from x , then both activations, then the output contributions h_1 and $-2h_2$, and finally add bias 1. A trace table makes an omitted bias or a sign error easier to locate than a final answer alone.

$x \cdot z_1$ and $z_2 \cdot h_1$ and $h_2 \cdot y$

x : 0

z_1 and z_2 : -1 and 2

h_1 and h_2 : 0 and 2

y : -3

x : 1

z_1 and z_2 : 1 and 1

h_1 and h_2 : 1 and 1

y : 0

x : 3

z_1 and z_2 : 5 and -1

h_1 and h_2 : 5 and 0

y : 6

At $x = 1$ the output is exactly zero. If we later declare class 1 for $y \geq 0$, equality belongs to class 1. Applying ReLU in the hidden layer does not make the final output nonnegative.

List the given facts and one fact that is not supplied.

3. See how it works

For $x = 0$, the first score is -1 , which ReLU changes to 0 . The second score is 2 , which ReLU leaves at 2 . Therefore $y = 0 - 4 + 1 = -3$. Negative output is allowed because this output layer uses identity.

```
def trace(x):  
    z = [2 * x - 1, -x + 2]  
    h = [max(0, value) for value in z]  
    y = h[0] - 2 * h[1] + 1  
    return z, h, y  
  
for x in [0, 1, 3]:  
    print(x, trace(x))
```

This optional Python uses only numeric inputs. Compare every returned component against the table. The same code being run twice is a repeatability check; comparing it with a separately calculated expectation is a stronger check of correctness.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

At $x = 3$ a trace gives $z = [5, -1]$, $h = [5, 0]$ and $y = 5$. Identify the first incorrect stage and repair it.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: A submitted $x = 0$ trace uses $h = -1$ and $y = 5$.

Prediction: Its first error is retaining the negative value after ReLU; correct $h = 0$ gives $y = 4$.

Reason: $\text{ReLU}(-1) = 0$, so the output must use 0, not the earlier preactivation.

5. Words that unlock the idea

Word · Meaning

Word: Trace

Meaning: A record of intermediate values in a computation

Word: Preactivation

Meaning: A weighted score before an activation function

Word: Dependency

Meaning: An earlier value needed to compute a later value

Word: Regression check

Meaning: A retained test rerun after a change to detect a new error

Use two terms to explain a step in this week's example.

6. Follow a worked example

Correct a trace from the earliest error

Trace stage at x is 0 · Expected value · Submitted value

Trace stage at x is 0: Scores

Expected value: [-1, 2]

Submitted value: [-1, 2]

Trace stage at x is 0: ReLU activations

Expected value: [0, 2]

Submitted value: [-1, 2]

Trace stage at x is 0: Output

Expected value: -3

Submitted value: -4

The columns compare expected and submitted traces. The first disagreement is the activation of -1, not the final addition. Recompute all dependent values after repairing it.

Trace x = 2, recording both scores, both activations and y. What happens to the second unit at its zero-score boundary?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Forward passes by hand

Audit a forward pass by writing every value at its correct stage.

The facts we will use

Toy scalar network $z=2x-1$, $h=\text{ReLU}(z)$, $y=-h+4$. Trace $x=0$ and $x=2$.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Forward passes by hand

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Preactivation

At $x=0$, $z=-1$; at $x=2$, $z=3$.



2. Activation

At $x=0$, $h=0$; at $x=2$, $h=3$.



3. Output

Outputs are 4 and 1 respectively.

CHANGE ONE THING

A submitted $x = 0$ trace uses $h=-1$ and $y = 5$.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Preactivation

What happens and why: At $x=0$, $z=-1$; at $x=2$, $z=3$.

Step: 2. Activation

What happens and why: At $x=0$, $h=0$; at $x=2$, $h=3$.

Step: 3. Output

What happens and why: Outputs are 4 and 1 respectively.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A forward pass applies a fixed network to an input; it does not update weights. Keep the preactivation z , activated value h and final output y in separate trace columns. For zero, ReLU changes minus one to zero. Using the earlier minus one in the output layer is the first error, even if later arithmetic follows that mistaken value correctly.

What this example does not establish

A correct forward trace verifies this declared computation; it does not show that the parameter choices solve a useful task.

7. Find and repair a mistake

A test where both scores are positive cannot reveal a missing ReLU. Include $x = 0$, where the first score is negative, and $x = 3$, where the second score is negative. Include $x = 0.5$ and $x = 2$ to exercise each hidden zero-score boundary. Neither a few examples nor an attractive diagram proves every implementation case is correct.

A missing output bias shifts every output by -1. A swapped hidden order may pass a case where h_1 equals h_2 ; $x = 1$ has this limitation. Keep several checks with different intermediate values, and retain them when repairing code.

Locate the earliest unsupported step

Trace $x = 0.5$ and $x = 2$. Which hidden score is zero in each case?

State one repair and the checks you would rerun.

8. Practise together

Trace $x = -1$ through the lesson network. Record z , h and y .

Trace $x = 2$, recording both scores, both activations and y . What happens to the second unit at its zero-score boundary?

Explain your prediction to a partner before comparing results.

Practise with less help: Locate a bias omission

Use workbook W2 for the connected example “Forward passes by hand”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Forward passes by hand. Proposed explanation: “ReLU means the output cannot be negative”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why should feedback identify h before correcting only y ? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

At $x = 3$ a trace gives $z = [5, -1]$, $h = [5, 0]$ and $y = 5$. Identify the first incorrect stage and repair it.

```
def trace(x):  
    z = [2 * x - 1, -x + 2]  
    h = [max(0, value) for value in z]  
    y = h[0] - 2 * h[1] + 1  
    return z, h, y  
  
for x in [0, 1, 3]:  
    print(x, trace(x))
```

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

Trace $x = -1$ through the lesson network. Record z , h and y .

Trace $x = 0.5$ and $x = 2$. Which hidden score is zero in each case?

Explain why testing only $x = 1$ in the lesson model may miss a hidden-order swap.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled forward passes by hand evidence sheet to Image-pattern Network Investigation. Show how you can record scores and activations in dependency order. Use the supplied fictional case and keep its

inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

A forward trace verifies what specified parameters calculate for one input. It does not show how those parameters were learned or whether the output is useful. Classification accuracy also needs a declared class rule, true labels and suitable independent examples.

A program matches y at $x = 1$ but its hidden activations are wrong. Is the implementation verified? Explain what evidence to retain after fixing it.

Next week separates the network architecture from its numerical parameters and counts the values that training could change.

Explain why testing only $x = 1$ in the lesson model may miss a hidden-order swap.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Correct a trace from the earliest error

Trace stage at x is 0 · Expected value · Submitted value

Trace stage at x is 0: Scores

Expected value: [-1, 2]

Submitted value: [-1, 2]

Trace stage at x is 0: ReLU activations

Expected value: [0, 2]

Submitted value: [-1, 2]

Trace stage at x is 0: Output

Expected value: -3

Submitted value: -4

The columns compare expected and submitted traces. The first disagreement is the activation of -1, not the final addition. Recompute all dependent values after repairing it.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Audit a different network independently

Use $z_1 = x + 1$, $z_2 = 2 - 2x$, $h = \text{ReLU}(z)$ component by component, and $y = 2h_1 + h_2 - 3$. Trace $x = 0$ and $x = 2$. A submitted $x = 2$ trace uses $h = [3, -2]$ and $y = 1$. Find its first error and correct all later values.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Use $z_1 = x + 1$, $z_2 = 2 - 2x$, $h = \text{ReLU}(z)$ component by component, and $y = 2h_1 + h_2 - 3$. Trace $x = 0$ and $x = 2$. A submitted $x = 2$ trace uses $h = [3, -2]$ and $y = 1$. Find its first error and correct all later values.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.